Structures and Materials

Tutorial | Dr.-Ing. Andreas Hermanutz

Elementary Concepts of Elasticity

1. Strain and Stress

Task 1 : The following displacement fields are generated in an aluminium plate (plane stress state) as a result of the external loading:

$$u_1 = (a_1 x_1 + a_0) x_2 \qquad u_2 = a_3 x_1 x_2$$

A) Find the engineering strain at position $x_1 = 2$ and $x_2 = 2$

$$\frac{\partial u_1}{\partial x_1} = a_1 x_2 \qquad \frac{\partial u_2}{\partial x_1} = a_3 x_2$$

$$\frac{\partial u_1}{\partial x_2} = a_1 x_1 + a_0 \qquad \frac{\partial u_2}{\partial x_2} = a_3 x_1$$

$$\varepsilon_{ij} = \begin{bmatrix} a_1 x_2 & \frac{1}{2}(a_1 x_1 + a_0 + a_3 x_2) \\ \frac{1}{2}(a_1 x_1 + a_0 + a_3 x_2) & a_3 x_1 \end{bmatrix}$$

$$\varepsilon_{ij}(2,2) = \begin{bmatrix} 2a_1 & \frac{1}{2}a_0 + a_1 + a_3 \\ \frac{1}{2}a_0 + a_1 + a_3 & 2a_3 \end{bmatrix}$$

1. Strain and Stress

B) Find the stress at position $x_1 = 2$ and $x_2 = 2$, for the aluminium plate $(E = 70GPa, \nu = 0.3)$

$$\boldsymbol{\sigma_{xx}} = \frac{\boldsymbol{E}}{\boldsymbol{1-\nu^2}}\left(\boldsymbol{\varepsilon_{xx}} + \boldsymbol{\nu\varepsilon_{yy}}\right) \qquad \boldsymbol{\sigma_{yy}} = \frac{\boldsymbol{E}}{\boldsymbol{1-\nu^2}}\left(\boldsymbol{\nu\varepsilon_{xx}} + \boldsymbol{\varepsilon_{yy}}\right) \qquad \boldsymbol{\sigma_{xy}} = \frac{\boldsymbol{E}}{\boldsymbol{1+\nu}}\left(\boldsymbol{\varepsilon_{xy}}\right)$$

$$\sigma_{xx} = \frac{70\ GPa}{1-0.3^2}(2a_1 + 0.6\ a_3) \qquad \sigma_{yy} = \frac{70\ GPa}{1-0.3^2}(0.6\ a_1 + a_3) \qquad \boldsymbol{\sigma_{xy}} = \frac{\boldsymbol{70GPa}}{\boldsymbol{1.3}}\left(\frac{1}{2}\boldsymbol{a_0} + \boldsymbol{a_1} + \boldsymbol{a_3}\right)$$

$$\boldsymbol{\sigma} = \begin{bmatrix} \dfrac{70\ GPa}{1-0.3^2}(2a_1 + 0.6\ a_3) & \dfrac{70GPa}{1.3}\left(\dfrac{1}{2}a_0 + a_1 + a_3\right) \\ \dfrac{70GPa}{1.3}\left(\dfrac{1}{2}a_0 + a_1 + a_3\right) & \dfrac{70\ GPa}{1-0.3^2}(0.6\ a_1 + a_3) \end{bmatrix}$$

$$S = \frac{1}{E}\begin{bmatrix} 1 & -\nu & 0 \\ -\nu & 1 & 0 \\ 0 & 0 & 2(1+\nu) \end{bmatrix}$$

$$Q = S^{-1} = \begin{bmatrix} \dfrac{E}{1-\nu^2} & \dfrac{E\nu}{1-\nu^2} & 0 \\ \dfrac{E\nu}{1-\nu^2} & \dfrac{E}{1-\nu^2} & 0 \\ 0 & 0 & \dfrac{1}{2(1+\nu)} \end{bmatrix}$$

$$\begin{Bmatrix} \boldsymbol{\sigma_{xx}} \\ \boldsymbol{\sigma_{yy}} \\ \boldsymbol{\sigma_{xy}} \end{Bmatrix} = \begin{bmatrix} \dfrac{E}{1-\nu^2} & \dfrac{E\nu}{1-\nu^2} & 0 \\ \dfrac{E\nu}{1-\nu^2} & \dfrac{E}{1-\nu^2} & 0 \\ 0 & 0 & \dfrac{E}{2(1+\nu)} \end{bmatrix} \begin{Bmatrix} \varepsilon_{xx} \\ \varepsilon_{yy} \\ 2\varepsilon_{xy} \end{Bmatrix}$$

2. Strain and Stress

Task 2: Given is a plane stress state defined by the stress tensor σ with constants a through d. What criteria must be satisfied for the state to be physically realizable??

$$\sigma = \begin{bmatrix} ax + by & ex - ay \\ ex - ay & cx + d \end{bmatrix}$$

The stress state must be in equilibrium with the applied body forces and surface tractions.

$$div(\sigma) + f = 0 \quad \Rightarrow \quad \frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} = 0 \qquad \frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} = 0$$

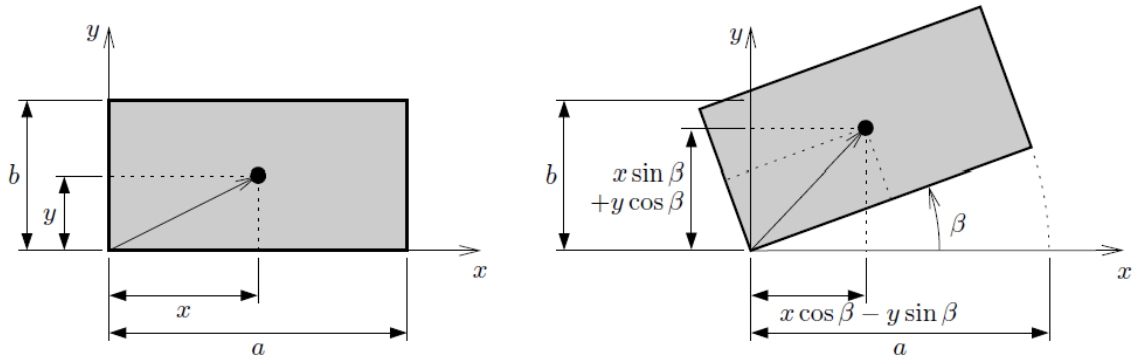
$$\frac{\partial \sigma_{xx}}{\partial x} = a \qquad \frac{\partial \tau_{xy}}{\partial y} = -a$$

$$\frac{\partial \sigma_{yy}}{\partial y} = 0 \qquad \frac{\partial \tau_{xy}}{\partial x} = e \quad \Rightarrow \quad e = 0$$

3. Strain and Stress

Task 3: An aluminium plate is rotated by an angle of $\beta = 90$

A) Define the displacement field.
B) Calculate the resulting strains.



Displacement field: $\vec{u} = \vec{x} - \vec{X}$

$u_1 = x\cos(\beta) - y\sin(\beta) - x$ $u_2 = x\sin(\beta) + y\cos(\beta) - y$

Strain field:

$\frac{\partial u_1}{\partial x} = \cos(\beta) - 1$ $\frac{\partial u_1}{\partial y} = -sin(\beta)$

$\frac{\partial u_2}{\partial x} = sin(\beta)$ $\frac{\partial u_2}{\partial y} = cos(\beta) - 1$

$$\varepsilon_{ij} = \begin{bmatrix} \cos(\beta) - 1 & 1/2(sin(\beta) - sin(\beta)) \\ 1/2(sin(\beta) - sin(\beta)) & cos(\beta) - 1 \end{bmatrix}$$

$$\varepsilon_{ij} = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$$

3. Strain and Stress

C) Compare the strains using the definition after Green Lagrange:

$$\varepsilon_{xx} = \frac{\partial u}{\partial x} + \frac{\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial v}{\partial x}\right)^2}{2} \qquad \varepsilon_{yy} = \frac{\partial v}{\partial y} + \frac{\left(\frac{\partial u}{\partial y}\right)^2 + \left(\frac{\partial v}{\partial y}\right)^2}{2} \qquad \varepsilon_{xy} = \frac{\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} + \frac{\partial u}{\partial x}\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}\frac{\partial v}{\partial y}}{2}$$

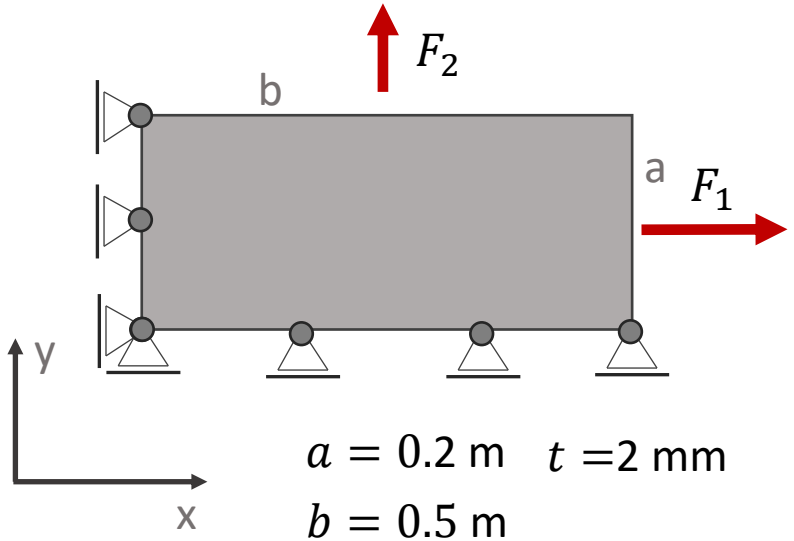
Green Lagrange Strain Tensor

$$\varepsilon_{xx} = \cos(\beta) - 1 + \frac{(\cos(\beta) - 1)^2 + (sin(\beta))^2}{2} = \cos(\beta) - 1 + \frac{cos^2(\beta) - 2\cos(\beta) + 1 + sin^2(\beta)}{2} = \cos(\beta) - 1 + \frac{2 - 2cos(\beta)}{2} = 0$$

$\varepsilon_{yy} = 0$

$\varepsilon_{xy} = 0$

4. Strain and Stress



$a = 0.2$ m $\quad t = 2$ mm

$b = 0.5$ m

Task 4: A rectangular aluminium plate ($E = 70$ GP$a, \nu = 0.3$) of dimensions a × b × t is given. The plate is constrained to move freely in the y-direction along its left edge and in the x-direction along its bottom edge. The plate is loaded by the force $F_1 = 25$kN and $F_2 = 10$kN, distributed uniformly along the edges.

A) Calculate the strains in the X and Y direction.

B) Calculate the displacement of the right edge in the X direction.

A)

$$\sigma_x = \frac{F_1}{a \cdot t}$$

$$\sigma_y = \frac{F_2}{b \cdot t}$$

$$\sigma_{xy} = 0.0 \text{ MPa}$$

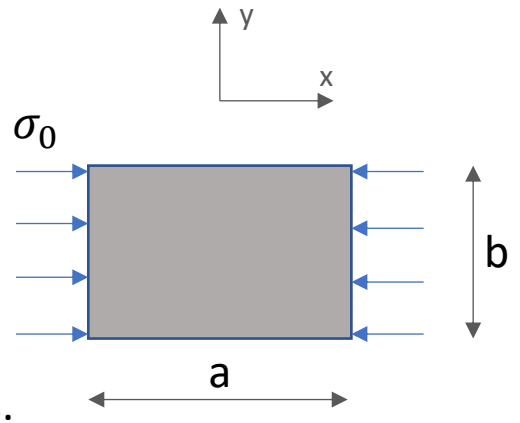
$$\begin{Bmatrix} \varepsilon_x \\ \varepsilon_y \\ \varepsilon_{xy} \end{Bmatrix} = \frac{1}{E} \begin{bmatrix} 1 & -\nu & 0 \\ -\nu & 1 & 0 \\ 0 & 0 & 2(1+\nu) \end{bmatrix} \begin{Bmatrix} \sigma_x \\ \sigma_y \\ \sigma_{xy} \end{Bmatrix} = \begin{Bmatrix} 0.00085 \\ -0.000125 \\ 0 \end{Bmatrix}$$

B)

$$\epsilon_x = \frac{du_x}{dx} \rightarrow u_x = \epsilon_x dx$$

$$\int_0^b \epsilon_x dx = u_x = \epsilon_x \cdot b = 0.425 \text{ mm}$$

5. Strain and Stress



Task 5: Consider a linear elastic isotropic plate of dimensions a×b. The plate's material properties are defined by the Young's modulus E and Poisson's ratio ν. The applied load is a uniform uniaxial compressive stress of magnitude σ_0 in the x-direction.

A) Analyze the stress and strain state a condition of free lateral expansion in the x and y directions.
B) How does the result change when expansion is constrained in the y-direction?

A)

$$\begin{Bmatrix} \epsilon_{xx} \\ \epsilon_{yy} \\ 2\epsilon_{xy} \end{Bmatrix} = \frac{1}{E}\begin{bmatrix} 1 & -\nu & 0 \\ -\nu & 1 & 0 \\ 0 & 0 & 2(1+\nu) \end{bmatrix}\begin{Bmatrix} -\sigma_0 \\ 0 \\ 0 \end{Bmatrix} = \begin{Bmatrix} -\frac{\sigma_0}{E} \\ \frac{\nu\sigma_0}{E} \\ 0 \end{Bmatrix}$$

B)

$$\epsilon_{yy} = \frac{1}{E}\left[\sigma_{yy} - \nu\sigma_{xx}\right] = 0 \qquad \sigma_{yy} = -\nu\sigma_0$$

$$\epsilon_{xx} = \frac{1}{E}\left[\sigma_{xx} - \nu\sigma_{yy}\right] = \frac{\sigma_0}{E}(\nu^2 - 1)$$